ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

TASK 9 – IMPLEMENT AND TEST THE MILK PAYMENT CALCULATOR

1. Aim

To implement and test a Python program that calculates the milk rate per litre and the total payment based on milk quantity, fat percentage, and SNF percentage.

2. Program Description

The program uses Python to calculate the payment for milk supplied by a farmer. The base rate is ₹30.00 per litre for milk containing 3.5% fat and 8.5% SNF. For every 0.1% increase or decrease in fat, the rate changes by ₹0.40. For every 0.1% increase or decrease in SNF, the rate changes by ₹0.30.

The program uses the Decimal data type for accurate financial calculations. A function named calculate_payment() calculates the rate per litre and the total payment amount.

3. Input Details

Farmer ID: F001

Milk Quantity: 10.5 litres

Fat Percentage: 4.0%

SNF Percentage: 8.7%

4. Calculation

Base Rate = ₹30.00

Fat Difference = 4.0 − 3.5 = 0.5%

Fat Adjustment = 5 × ₹0.40 = ₹2.00

SNF Difference = 8.7 − 8.5 = 0.2%

SNF Adjustment = 2 × ₹0.30 = ₹0.60

Rate Per Litre = ₹30.00 + ₹2.00 + ₹0.60 = ₹32.60

Total Payment = 10.5 × ₹32.60 = ₹342.30

5. Output

Farmer ID: F001

Milk Quantity: 10.5 litres

Fat: 4.0%

SNF: 8.7%

Rate Per Litre: ₹32.60

Payment Amount: ₹342.30

6. Testing and Validation

The program was executed using the given input values. The calculated rate per litre was ₹32.60, and the total payment was ₹342.30. The output matches the expected calculation.

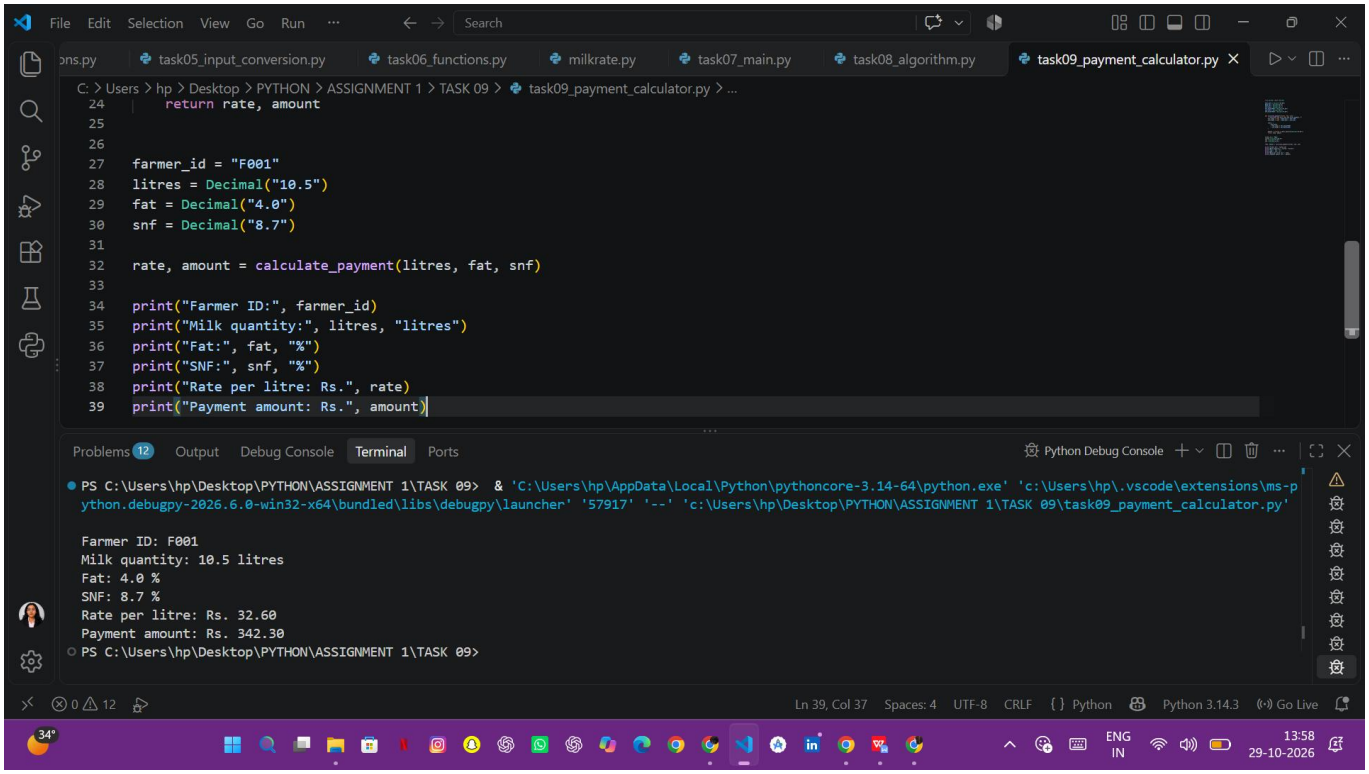
The Decimal data type helps avoid common floating-point precision problems in financial calculations. The function returns the rate and payment amount, making the program easier to test and reuse.

7. Time and Space Complexity

The program performs a fixed number of calculations for one farmer. Therefore, the time complexity is O(1), and the auxiliary space complexity is O(1).

8. Result

The milk payment calculator was successfully implemented and tested in Python. It correctly calculates the rate per litre and total payment based on milk quantity, fat percentage, and SNF percentage.


File Edit Selection View Go Run
Search
task05_input_conversion.py
task06_functions.py
milkrate.py
task07_main.py
task08_algorithm.py
task09_payment_calculator.py
C: > Users > hp > Desktop > PYTHON > ASSIGNMENT 1 > TASK 09 > task09_payment_calculator.py > ...
return rate, amount
farmer_id = "F001"
litres = Decimal("10.5")
fat = Decimal("4.0")
snf = Decimal("8.7")
rate, amount = calculate_payment(litres, fat, snf)
print("Farmer ID:", farmer_id)
print("Milk quantity:", litres, "litres")
print("Fat:", fat, "%")
print("SNF:", snf, "%")
print("Rate per litre: Rs.", rate)
print("Payment amount: Rs.", amount)
Problems 12 Output Debug Console Terminal Ports
Python Debug Console
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 09> & 'C:\Users\hp\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '57917' '--' 'c:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 09\task09_payment_calculator.py'
Farmer ID: F001
Milk quantity: 10.5 litres
Fat: 4.0 %
SNF: 8.7 %
Rate per litre: Rs. 32.60
Payment amount: Rs. 342.30
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 09>
Ln 39, Col 37 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN
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29-10-2026